

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Properties of pollutants & environmental factors - Set 1

Mark scheme

AQA spec 3.4.1-3.4.2; Pollution; textbook pp. 236-242

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• The time taken for half of a pollutant (released into the environment) to break down / degrade	1
01.2	• DDT is liposoluble so it is stored in fat / can pass through phospholipid cell membranes and is not excreted (nitrates are water soluble and excreted easily) • DDT is persistent (long half-life) so it is not broken down before it is passed to the next trophic level • So DDT bioaccumulates in each organism and becomes more concentrated at each successive trophic level	3
01.3	<i>Any 2 from:</i> • The gas is denser than air so it settled close to the ground / did not rise and disperse • Low wind velocity (at night) meant it was not dispersed / diluted • Short half-life so it broke down before travelling far	2
01.4	• High solubility means nitrates are easily dispersed / leached / carried by water so they affect a larger area (eg rivers, aquifers, the sea) • But dispersal dilutes the pollutant so the concentration at any one point may be lower	2
01.5	<i>Any 2 from:</i> • Noise / sound • Heat / thermal pollution • Ionising radiation • Light	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	

Question	Marking guidance	Marks
02.1	• Mutagen: an agent that changes the chemical structure / base sequence of DNA (causes mutations), eg ionising radiation, UV, PCBs, cadmium, asbestos • Carcinogen: a mutagen that causes cancer / uncontrolled cell division producing a tumour • Teratogen: an agent that causes birth abnormalities by preventing normal gene expression (inhibiting proteins / enzymes) without changing DNA structure, eg mercury, 2,4,5-T	3
02.2	• Teratogens do not change the DNA structure, so the abnormality is not in the genes and cannot be passed on • Mutagens change DNA; if the mutation is in a gamete / egg / sperm (gonadic effect) it can be inherited	2
02.3	• Somatic: mutation in a general body cell, affecting the individual only (eg cancer / tumour) • Gonadic: mutation in cells of the ovaries or testes / gametes / embryo, which may cause abnormalities in offspring	2
02.4	• Two or more pollutants whose effects interact to produce a different (usually more serious) effect • The pollutants themselves do not react to form a new pollutant (that would be a secondary pollutant) • Example: ozone damages leaf cuticles so sulfur dioxide can cause more damage to the exposed cells; smoke and sulfur dioxide in smogs; neonicotinoids with fungicides	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 239-240; AQA spec 3.4.1.	
03.1	• Temperature inversion	1
03.2	• On night A, temperature falls with altitude; warm effluent gases are less dense / more buoyant than the surrounding air so they rise • As they rise they cool but the surrounding air is also cooler so they stay buoyant and continue to rise, dispersing and diluting • On night B the air near the ground is colder / denser than the air above (inversion layer at about 400 m) • Pollutants cool as they rise and become denser than the warmer air above so they are trapped below the inversion layer and become concentrated	4
03.3	<i>Any 3 from:</i> • Cloudless sky allows infrared energy to be radiated from the ground so the ground cools rapidly • The cold ground cools the air in contact with it, producing a cold dense layer near the ground • Low wind velocity means the air layers of different temperatures do not mix • Valley allows cold dense air to collect / drain into the low ground	3
03.4	<i>Any 2 from:</i> • Smoke smogs (eg London 1952) • Photochemical smogs (eg Los Angeles, Mexico City) • Build-up of NO_x / carbon monoxide / PM₁₀ in cities	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 241-242; AQA spec 3.4.2.2.	

Question	Marking guidance	Marks
04.1	• Temperature: most chemical / biological reactions are faster at higher temperature so degradable pollutants (eg sewage) break down faster; but faster decomposition may increase deoxygenation • Light: provides the activation energy for chemical reactions; photodegradation of pesticides (eg neonicotinoids break down fastest in sunlight); light drives photochemical smog reactions producing secondary pollutants • Oxygen: needed for aerobic bacterial decomposition of organic wastes / sewage; oxidation of sulfide ores produces sulfur dioxide / acid mine drainage; oxidation state affects the solubility of metals	6
04.2	<i>Any 2 from:</i> • Pollutant ions adsorb onto (the surface of) clay particles / organic matter in soil or sediment • So they are immobilised and cannot be absorbed by organisms / leached into water; radiation is absorbed in the ground • But they may be released later if conditions change (eg storms disturbing sediment, low pH)	2
04.3	• Many heavy metals (eg lead, aluminium) are more soluble at low pH • So they are mobilised / leached from soil or sediments and can be absorbed by organisms / enter water bodies	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 240-242, 255; AQA spec 3.4.2.2.	
05.1	• $(4.8 - 1.4) / 4.8 \times 100$ • = 70.8% (accept 71%)	2
05.2	• A tributary joins the river, more than doubling the flow ($15 \text{ to } 36 \text{ m}^3 \text{ s}^{-1}$) • So the pollutant is diluted by the extra water (dispersal), not broken down	2
05.3	• The pollutant is persistent so it is not degraded / broken down as it travels • Little extra water joins the river (flow 36 to 38) so little further dilution	2
05.4	<i>Any 2 from:</i> • Lake has a long residence time / water does not flow away so the pollutant accumulates rather than being carried away • Smaller volume of water / less dilution / less dispersal by currents • Persistent pollutant gradually becomes more concentrated (eg by evaporation)	2
05.5	• Marine discharge: large volume of water and currents dilute and disperse the pollutant to acceptably low concentrations (control of emission location) • River: small volume so concentrations stay high / drinking water abstraction / sensitive wildlife downstream	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-241, 254-255; AQA spec 3.4.1, 3.4.2.1.	
06.1	• B - Tropospheric ozone (formed by reactions of NO_x in sunlight)	1

Question	Marking guidance	Marks
06.2	• Primary pollutant: released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores; NO_x from high-temperature combustion • Secondary pollutant: produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) • Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid; NO_x dissolve to form nitric acid • The acids (sulfurous, sulfuric, nitric) are secondary pollutants; ozone itself is a secondary pollutant of NO_x photochemistry	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	

Question	Marking guidance	Marks
07.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • State of matter: solids deposit near the source, gases disperse widely in the atmosphere; density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel • Persistence / half-life: persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution; non-persistent pollutants (sewage, organophosphates) can be allowed to degrade • Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk; LD50 / MDAF / MDNF used to set doses • Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) • Reactivity: reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants • Adsorption onto clay / sediments immobilises metals and caesium; can be used deliberately (activated carbon filters, polymer adsorption) • Mutagenic / carcinogenic / teratogenic action means no safe dose; exposure kept As Low As Reasonably Achievable (ALARA) • Synergism (ozone + SO₂; smoke + SO₂) means pollutants must be considered together • Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring • Evaluation: properties of new chemicals can be predicted from similar ones (precautionary principle); understanding allows prevention rather than clean-up	9
07.2	• The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
08.1	• Inorganic mercury sank into the sediments / deep water where conditions are anaerobic / low oxygen • It was chemically reduced / converted by (anaerobic) bacteria to methyl mercury	2

Question	Marking guidance	Marks
08.2	• Methyl mercury is liposoluble so it passes through cell membranes and is stored in fat / not excreted • So it bioaccumulates in organisms and biomagnifies along the food chain (eg in fish, then humans eating fish) • It is a neurotoxin / teratogen causing brain damage and birth abnormalities (eg Minamata)	3
08.3	• Mercury is persistent / not degraded; it is an element so cannot be broken down • It remains in the sediments and continues to be released / converted and taken up by the food chain; long-lived fish continue to accumulate it	2
08.4	<i>Any 3 from:</i> • Water current velocity and direction (dispersal) • Properties of the pollutant: solubility, persistence, liposolubility / potential to bioaccumulate • Food chain pathways to humans (fish, shellfish) / the critical group • Sediment type / adsorption onto clay • Volume of the water body / residence time / dilution • Sensitive locations downstream (fisheries, drinking water abstraction)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 238-242; AQA spec 3.4.1, 3.4.2.	